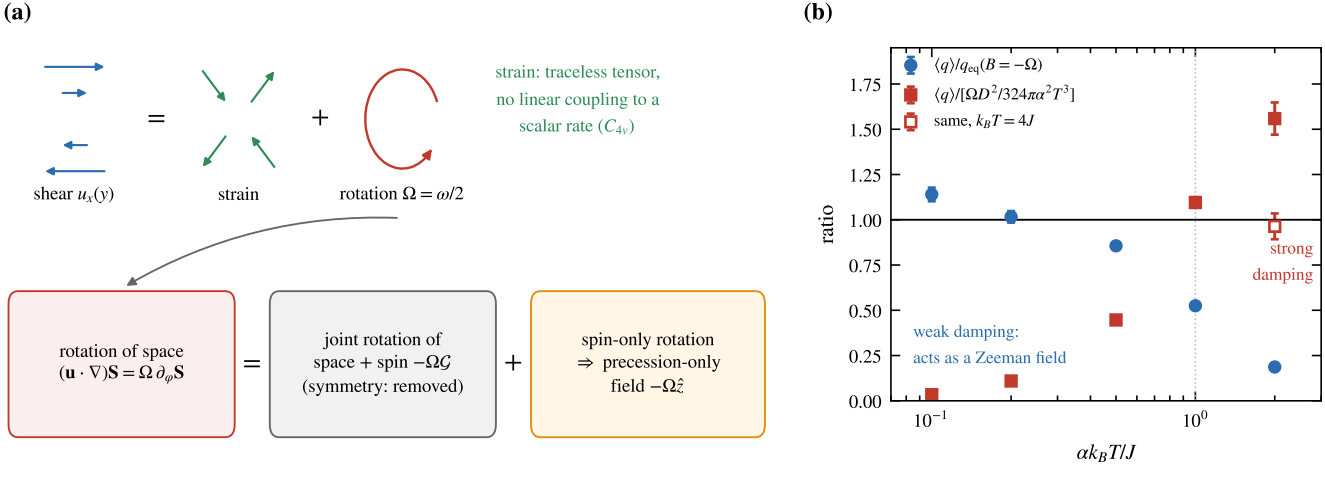
(a)
shear $u_x(y)$
=
strain
+
rotation $\Omega = \omega/2$
strain: traceless tensor,
no linear coupling to a
scalar rate (C_{4v})
rotation of space
$(\mathbf{u}\cdot\nabla)\mathbf{S} = \Omega\,\partial_\varphi \mathbf{S}$
=
joint rotation of
space + spin $-\Omega\mathcal{G}$
(symmetry: removed)
+
spin-only rotation
$\Rightarrow$ precession-only
field $-\Omega\hat{z}$
(b)
$\langle q\rangle/q_{\rm eq}(B=-\Omega)$
$\langle q\rangle/[\Omega D^2/324\pi\alpha^2T^3]$
same, $k_BT = 4J$
ratio
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
10^{-1}
10^{0}
$\alpha k_BT/J$
strong
damping
weak damping:
acts as a Zeeman field